

Zion R. Perry

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EDUCATION

Yale University, New Haven, CT 2020-2026

Ph.D., Molecular Biophysics and Biochemistry

Dissertation title: “Computational and Experimental Investigations of Noncoding RNA Structures”

Advisor: Anna Marie Pyle

Dissertation Committee: Matthew Simon (Chair), Karla Neugebauer, Sigrid Nachtergaele

Massachusetts Institute of Technology, Cambridge, MA 2016-2020

B.S., Biological Engineering, GPA: 4.7/5.0

RESEARCH INTERESTS & SKILLS

- Interests: Computational Structural Biology, Protein and RNA Folding and Design, Bioinformatics, Biochemistry
- Programming Languages: Python, C++, R, MATLAB, Bash, Wolfram/Mathematica
- Software & Tools: High Performance Computing, Git, Rosetta/PyRosetta, PyMOL, Jupyter, Bioinformatics Tools
- Experimental: Next-Generation Sequencing (SHAPE-MaP/DMS-MaP, MRT-ModSeq), Protein and RNA Purification, Biochemical Assays, Tissue Culture, Molecular Cloning, Light and Fluorescence Microscopy

RESEARCH EXPERIENCE

Graduate Researcher, Laboratory of Anna Marie Pyle Jun. 2021-May 2026

Department of Molecular, Cellular, and Developmental Biology, Yale University, New Haven, CT

- Led experimental project on the structure of a long noncoding RNA in breast cancer cells, processing NGS data in HPC clusters and integrating experimental results with Python/Jupyter frameworks for insights into secondary structure, protein binding, and RNA modifications in cells.
- Developed automated bioinformatics pipeline for discovery and alignment of RNA genes in 190 unannotated primate genomes, enabling comparative sequence and structural analyses of noncoding RNA targets.
- Developed C++ software for RNA structure prediction and refinement from coarse-grained structures with 353x faster runtime and 54% higher accuracy compared to state-of-the-art programs. Received Best Speaker Award for this work.

Computational Research Intern, Laboratory of Tanja Kortemme Jun.-Aug. 2019

Department of Bioengineering, University of California San Francisco, San Francisco, CA

- Developed novel protein binding site design pipeline in Rosetta that incorporates solvation, using Python, Bash, and HPC workflows. Developed PyMOL visualization method for quantifying hydrogen bonding propensity.
- Experimentally validated the function of top computational designs using an *in cellulo* reporter assay to inform successive design iterations.

Undergraduate Researcher, Laboratory of Tania A. Baker Jun. 2018-May 2019

Department of Biology, Massachusetts Institute of Technology, Cambridge, MA

- Designed and purified panel of bacterial proteases and substrates. Quantified kinetic differences across constructs using MATLAB degradation curve analysis, informing mechanism of protease-substrate recognition.

PUBLICATIONS

Published

- **Perry, Z. R.**, Pyle, A. M., & Zhang, C. (2023). Arena: Rapid and Accurate Reconstruction of Full Atomic RNA Structures from Coarse-grained Models. *Journal of molecular biology*, 435(18), 168210. <https://doi.org/10.1016/j.jmb.2023.168210>

- Mahadeshwar, G., Tavares, R. C. A., Wan, H., **Perry, Z. R.**, & Pyle, A. M. (2023). RScanner: rapid assessment and visualization of RNA structure content. *Bioinformatics*, 39(3), btad111. <https://doi.org/10.1093/bioinformatics/btad111>
- Glasgow, A., Hobbs, H. T., **Perry, Z. R.**, Wells, M. L., Marqusee, S., & Kortemme, T. (2023). Ligand-specific changes in conformational flexibility mediate long-range allostery in the lac repressor. *Nature communications*, 14(1), 1179. <https://doi.org/10.1038/s41467-023-36798-1>

Under Revision/Preprint

- **Perry, Z. R.**, Beeram, A., Lee, L., Pyle, A. M. (2026). Conserved structural features of the lncRNA HOTAIR in breast cancer cells. *BioRxiv*. <https://doi.org/10.64898/2026.05.18.725451> [Under revision at *RNA*]
- Beeram, A., **Perry, Z. R.**, Pyle, A. M. (2026). Sequence alignment of the primate lineage reveals evolutionary divergence and conserved secondary structural motifs in noncoding RNAs. *BioRxiv*. <https://doi.org/10.64898/2026.05.18.725462> [Under revision at *RNA*]
- Xiao, W., Hann, T., **Perry, Z. R.**, Pandey, V., Cheng, J., Wohlschlegel, J., Pyle, A. M., Black, D. (2026). The long noncoding RNA Malat1 contains an internal ribosome entry site mediating micropeptide translation. *BioRxiv*. <https://doi.org/10.64898/2026.02.11.705401> [Under revision at *Science Advances*]
- Kumar, A., Wan, H., **Perry, Z. R.**, Patel, S., Tavares, R., Pyle, A. M. (2023). Discovery of a Well-Folded Protein Interaction Hub Within the Human Long Non-Coding RNA *NORAD*. *BioRxiv*. <https://doi.org/10.1101/2023.08.07.552337>

GRANTS & FELLOWSHIPS

National Science Foundation Conference Grant	\$15,000	2025
McDougal Graduate Teaching Fellowship	\$11,300	2023-2025
National Science Foundation Graduate Research Fellowship Program	\$102,000	2020-2025

HONORS & AWARDS

Certificate of College Teaching Preparation, Yale	2025
Certificate in Physical and Engineering Biology, Yale	2024
Biophysics and Structural Biology Symposium Best Speaker Award, Yale	2023
Excellence in Teaching Award, Yale	2022
The Protein Society Diversity Award	2021
Gordon Engineering Leadership Program, MIT	2019
Johnson & Johnson Undergraduate Research Scholar, MIT	2018

TEACHING & MENTORSHIP EXPERIENCE

Research Mentorship

- Logan Lee (Undergraduate researcher) – Noncoding RNA bioinformatics Feb. 2026-July 2026
- Anish Beeram (Undergraduate researcher) – Noncoding RNA structure and bioinformatics May 2025-May 2026
- Natalia Bertolotti (Graduate rotation student) – lncRNA regulation of gene expression Jan. 2024-Mar. 2024

McDougal Teaching Fellow

Sept. 2023-May 2025

Poorvu Center for Teaching and Learning, Yale University, New Haven, CT

- Facilitated 11 interactive pedagogical workshops to equip graduate students and postdocs on best teaching practices.
- Provided classroom consultations to 5 graduate students, providing personalized feedback to enhance teaching.

Graduate Teaching Fellow

Sept.-Dec. 2021 & 2022

Department of Molecular Biophysics and Biochemistry, Yale University, New Haven, CT

Course: Biology at the Molecular Level

- Designed curriculum components (weekly lessons, homework, exams, labs), held weekly office hours, and led weekly section of ~15 students on coding and data analysis in Mathematica.
- Received student ratings of 5.0/5.0 (n=9) in 2021, 4.7/5.0 (n=16) in 2022, and 2022 Excellence in Teaching Award.

Science Instructor and Curriculum Designer

Summers 2023-2025

Homeward Youth Camp, New York, NY

Courses: Beautiful Biology (Computational Biology, Microscopy and Cell Biology), Molecular Biology of the Senses, Forensics, Professional Development

- Designed and taught 43 hours of interactive classes for international high school students at a 3-week summer camp.

Laboratory Instructor

Jul. 2023

Yale Pathways to Science, New Haven, CT

Course: How do we detect infectious viruses?

- Mentored 12 high school students in laboratory setting, leading a “how to pipette” activity and assisting students with tissue culture, centrifugation, and fluorescent microscopy. Received student rating of 4.5/5 (n=12).

Undergraduate Teaching Assistant

Sept.-Dec. 2019

MIT Department of Biology, Cambridge, MA

Course: Cell Biology

- Led weekly section of ~20 sections, held weekly office hours, graded exams, and moderated online discussions.
- Received student rating of 7.0/7.0 (n=17).

Other

- Engineering Design Instructor, MIT Development Lab, Thailand Aug. 2018
- English and Math Instructor, Mamelodi Initiative, University of Pretoria, South Africa Jun.-Jul. 2017
- STEM Camp Mentor, DynaMIT Program, MIT, Cambridge, MA Aug. 2017

PROFESSIONAL SERVICE & MEMBERSHIPS

Conference Chair, Gordon Research Seminar on Nucleic Acids, Lucca, Italy Jun. 2025

- Led planning and execution of international conference for 40 early-career researchers, managing speaker invitations, reviewing applications, and curating scientific sessions. On the final survey, 83% of attendees indicated no improvements needed, highlighting engaging poster sessions and a collegial discussion environment.
- Secured \$15K in funding through successful National Science Foundation proposal, the highest awarded in the conference’s history. Managed conference fundraising and budget allocation.

Manuscript Reviewer

- PeerJ
- Journal of Molecular Biology

Professional Memberships

The RNA Society, 2022-2025

The Protein Society, 2021-2023

The Biophysical Society, 2020

PRESENTATIONS

Oral Presentations

- Perry, Z. R.**, Pyle, A.M. “HOTAIR” rises: Overcoming challenges in studying the long noncoding RNA HOTAIR. Physical and Engineering Biology Discussion Group (Nov. 2023) New Haven, CT.
- Perry, Z. R.**, Zhang, C., Pyle, A.M. Rapid and accurate reconstruction of full atomic RNA structures. Yale Biophysics and Structural Biology Symposium (May 2023) New Haven, CT.
- Perry, Z. R.**, Glasgow, A., Kortemme, T. Semi-explicit solvation improves ligand binding site design in an allosteric protein. University of California, San Francisco Summer Research Training Program (Jul. 2019) San Francisco, CA.

Poster Presentations

- Perry, Z. R.**, Pyle, A.M. Mapping structural complexity in a lncRNA. Gordon Research Seminar on Nucleic Acids (Jun. 2025) Lucca, IT.
- Perry, Z. R.**, Mahadeshwar, G., Tavares, R., Wan, H., Pyle, A.M. Structure of the long noncoding RNA HOTAIR in breast cancer. RNA Society (May 2024) Edinburgh, SCT.

- **Perry, Z. R.**, Kumar, A., Patel, S., Mahadeshwar, G., Tavares, R., Wan, H., Pyle, A.M. Structure and function investigations of long noncoding RNAs. Gordon Research Seminar and Conference on Nucleic Acids (Jun. 2023) Newry, ME.
- **Perry, Z. R.**, Hann, T., Luo, M., Tavares, R., Mahadeshwar, G., Wan, H., Pyle, A.M. RNA structure-guided investigations of long RNAs. RNA Society (Jun. 2022) Boulder, CO.
- **Perry, Z. R.**, Glasgow, A., Kortemme, T. Semi-explicit solvation improves ligand binding site design in an allosteric protein. The Protein Society (Jul. 2021) virtual; Biophysical Society Meeting (Feb. 2020) San Diego, CA; Rosetta Conference (Aug. 2019) Seattle, WA; University of California, San Francisco Summer Research Training Program (Jul. 2019) San Francisco, CA.
- **Perry, Z. R.**, Kim, S., Baker, T. Recognition of the Transcription Factor FixK₂ by the AAA+ Protease ClpAP in *Bradyrhizobium japonicum*. MIT Life Sciences Research Symposium (May 2019) Cambridge, MA.

REFERENCES

Research References

Anna Marie Pyle, Ph.D. – Dissertation Advisor

Professor of Molecular, Cellular, and Developmental Biology, Yale University, New Haven, CT

Email: anna.pyle@yale.edu

Matthew Simon, Ph.D. – Dissertation Committee Chair

Associate Professor of Molecular Biophysics and Biochemistry, Yale University, New Haven, CT

Email: matthew.simon@yale.edu

Anum Glasgow, Ph.D. – Computational Research Mentor

Assistant Professor of Biochemistry and Molecular Biophysics, Columbia University, New York, New York

Email: ag4522@cumc.columbia.edu

Chengxin Zhang, Ph.D. – Computational Research Mentor

Principle Investigator, Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, Shenzhen, China

Email: cx.zhang2@siat.ac.cn

Teaching References

Enrique M. De La Cruz, Ph.D. – Instructor of Course for Graduate Teaching Fellowship

Professor of Molecular Biophysics and Biochemistry, Yale University, New Haven, CT

Email: enrique.delacruz@yale.edu

Gina Marie Hurley, Ph.D. – Supervisor for McDougal Teaching Fellowship

Associate Director, Teaching Development and Initiatives, Yale University, New Haven, CT

Email: gina.hurley@yale.edu